# Domain-Aware Scaling Laws Uncover Data Synergy

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## Abstract

Machine learning progress is often attributed to scaling model size and dataset volume, yet the composition of data can be just as consequential. Empirical findings repeatedly show that combining datasets from different domains yields nontrivial interactions. For instance, adding code improves mathematical reasoning, while certain mixtures introduce interference that reduces model performance. We refer to these effects collectively as data *synergy*, where the contribution of multiple domains exceeds or falls short of the sum of their isolated contributions. In this work, we formalize and quantify data synergy in language model pretraining. Leveraging observational variation across open-weight LLMs with diverse pretraining mixtures, we estimate both direct domain-to-benchmark synergy (how one domain contributes to performance on another) and a second-order domain-domain synergy (capabilities that require co-occurrence of multiple domains). Our framework improves predictive accuracy over domain-agnostic scaling laws and recovers stable synergy estimates. We validate these estimates by training models on predicted optimal and predicted anti-optimal mixtures and confirm that our synergy estimates correctly predict performance rankings.

## 1 Introduction

Recent improvements in Large Language Models (LLMs) are strongly shaped by their pretraining data distribution (Li et al., 2024a; Soldaini et al., 2024), yet prior work abstracts away composition and reduces data to an undifferentiated token count (Kaplan et al., 2020; Hoffmann et al., 2022). In practice, however, pretraining corpora are mixtures of data from different domains (e.g., web, books, code, math) and small shifts in composition can significantly impact model capabilities (Ye et al., 2024; Liu et al., 2024). In particular, empirical studies repeatedly report interactions between data domains that are systematic rather than anecdotal: code pretraining can improve mathematical (Lu et al., 2024; Azerbayev et al., 2023) and logical (Yu et al., 2024) reasoning, math and code mixtures often outperform either alone (Aryabumi et al., 2024), while other combinations lead to interference and degrade performance on certain tasks (Zheng et al., 2024; Li et al., 2024b; Gu et al., 2024). These observations suggest that tokens are not interchangeable. What matters is not only how much data we train on, but also *what kinds of data are combined*.

We refer to these interactions collectively as **data synergy**. We distinguish two complementary forms (Figure 1). The first is *domain→benchmark synergy*: the extent to which pretraining data from one domain helps or hurts performance on a given benchmark, beyond what would be expected from total token count. The second is *domain-domain synergy*: non-additive effects that arise when two domains co-occur in the training mixture, so that their joint contribution is greater or smaller than the sum of their isolated effects. The former captures the effect of a training domain on a given benchmark or evaluation domain, and the latter captures higher-order complementarities or interference internal to the pretraining corpus itself.

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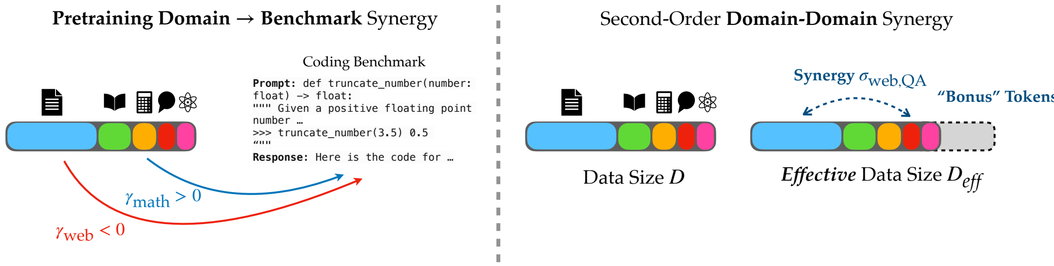


Figure 1: **Two forms of data synergy:** Pretraining corpora are mixtures of different data domains (Web, Books, Math, Q&A, Science, . . . ), and their *composition* shapes downstream performance. ***Left:*** Each domain modifies the *rate* at which additional data reduces loss on a given benchmark, which is captured by a per-domain adjustment $\gamma_{\text{domain}}$ to the data scaling exponent. On a coding benchmark, adding math data reduces loss faster, while web data reduces it more slowly. ***Right:*** When two synergistic pretraining domains co-occur, their pairwise interaction $\sigma_{kk'}$ contributes additional "bonus" tokens that vanish if either domain is absent, so the corpus behaves as if it had a larger *effective* data size $D_{eff} > D$. These synergies are shared across benchmarks, but their bonus tokens stay benchmark-specific.

Most existing approaches overlook data synergy: either they treat pretraining corpora as homogeneous, or they model composition without modeling the interactions between domains. Classical scaling laws, for instance, relate loss to parameter count and total data but are domain-agnostic, since they assume all tokens contribute equally (Kaplan et al., 2020; Hoffmann et al., 2022). Similarly, data attribution and curation methods usually learn per-example or per-domain contributions to a target metric (Bae et al., 2024; Grosse et al., 2023; Ilyas et al., 2022), and mixture optimization methods often search over weights under an implicit assumption of independent domains (Xie et al., 2023; Liu et al., 2024). What is missing is a formalization of data synergy that separates the aggregate benefit of more data from the effect of its composition, and quantifies when two domains together yield more (or less) than the sum of their parts.

In this paper, we present **domain-aware scaling laws**, a framework that quantifies and models data synergy in LLM pretraining. Rather than training a large controlled set of models, we leverage observational variation across open-weight LLMs with diverse pretraining mixtures. Our framework modifies standard scaling laws with additional terms that capture both first-order domain→benchmark synergy and second-order synergy between pretraining domains. The resulting estimators improve predictive accuracy over domain-agnostic scaling laws and recover sparse, interpretable estimates of synergy and interference. For example, we find recurring positive interactions between code and math, alongside negative interactions for some domain pairs that consistently degrade performance across different benchmarks. Our contributions are as follows:

- We provide a formal, quantifiable definition of data synergy that captures prior empirical observations.
- We introduce domain-aware scaling laws with first- and second-order synergy coefficients that improve predictive model performance over standard scaling laws.
- We recover stable, interpretable synergy estimates that provide actionable insights for data curation and acquisition.
- We validate the framework by training models on predicted-optimal and predicted-anti-optimal mixtures and show that observationally derived synergy estimates correctly predict performance rankings across multiple benchmarks.

## 2 Domain and Synergy-Aware Scaling Laws

In this section, we develop scaling laws that progressively incorporate training data composition and synergy into the standard scaling laws. We begin with the problem setup and a domain-agnostic baseline in Sections 2.1 and 2.2, then extend the data term to capture first-order domain→benchmark synergy in Section 2.3 and second-order synergy between pretraining domains in Section 2.4.

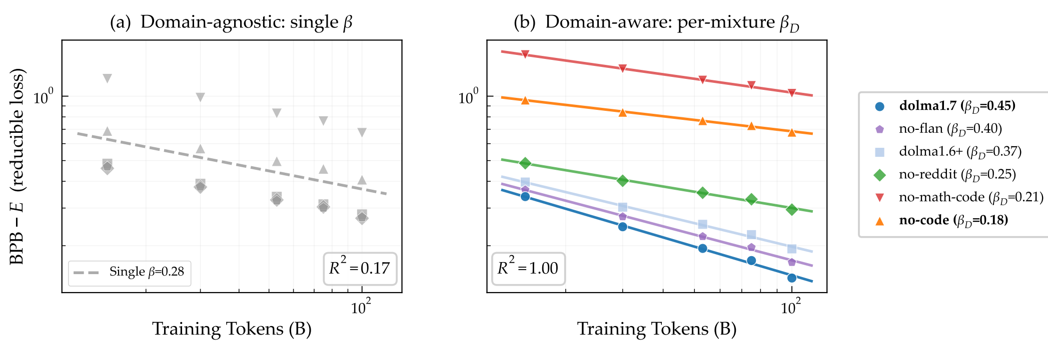


Figure 2: **Data scaling exponent depends on pretraining mixture**: a motivating example on `HumanEval`. ***(a)*** A single domain-agnostic scaling exponent ($\beta = 0.28$) fits the data poorly ($R^2 = 0.17$), as models with different mixtures show different reducible losses at a fixed token count. ***(b)*** Allowing a per-mixture exponent $\beta_D$ captures the variation from different data mixtures ($R^2 = 1.00$). Removing code or math data *reduces* the data scaling exponent (the slope here), while the full mixture scales faster.

### 2.1 Problem Setup

Let $\{M_1, \ldots, M_m\}$ denote a set of language models (e.g., open-weight LLMs on Huggingface), and let $\{D_1, \ldots, D_n\}$ be evaluation domains. For every model-domain pair we observe the loss value $L = [l_{i,j}] \in \mathbb{R}^{m \times n}$ where rows $L_{i,:}$ summarize the performance of model $M_i$ across domains and columns $L_{:,j}$ compare models on domain $D_j$. Alongside $L$ we collect statistics of model $M_i$, such as parameter count $N_i$, total pretraining tokens $d_i$, and composition (mixture shares) of the pretraining data. Let $u_{i,k} \in [0, 1]$ be the fraction of tokens from training domain $D_k$ that model $M_i$ is pretrained on, with $\sum_k u_{i,k} = 1$ and $d_i = \sum_k u_{i,k} d_i$.

Our goal is to quantify *data synergy across domains*, or how training on pretraining domain(s) affects loss on a benchmark, after accounting for model scale and total tokens.

### 2.2 Domain-agnostic scaling law

We begin with a domain-agnostic scaling law that explains loss variation using only model size $N$ in terms of number of parameters and total pretraining tokens $D$. This is the baseline model against which composition- and synergy-aware adjustments will be evaluated. Following Chinchilla scaling laws (Hoffmann et al., 2022), we have the parametric form

$$L(N, D) = L_\infty + AN^{-\alpha} + BD^{-\beta}, \tag{1}$$

where $L_\infty > 0$ is the irreducible loss, $A, B > 0$ are scale coefficients for parameter and data terms, and $\alpha, \beta > 0$ are the parameter and data scaling exponents. Typical values of the exponents are $\alpha \approx 0.06$ and $\beta \approx 0.20$.

In practice we fit this baseline *per benchmark j* and allow $(L_{\infty,j}, A_j, B_j, \alpha_j, \beta_j)$ to vary across evaluation domains, after transforming benchmark performance metrics to a common pseudo-loss scale. Let $s = \log N$ and $d = \log D$, and define $e_j = \log L_{\infty,j}$, $a_j = \log A_j$, and $b_j = \log B_j$. Writing (1) in log parameters gives the numerically-stable log-sum-exp (LSE) form:

$$\log L_j(s, d) = \mathrm{LSE}\big(e_j,\ a_j - \alpha_j s,\ b_j - \beta_j d\big), \tag{2}$$

where $\mathrm{LSE}(x_1, x_2, x_3) = \log\big(e^{x_1} + e^{x_2} + e^{x_3}\big)$. Equation (2) is our *domain-agnostic* baseline: the expected log-loss depends only on total parameters and total tokens. In the next subsections we expand and modify the *data* term $b - \beta \log D$ to account for the data mixture and to estimate synergies, while preserving the overall form of the scaling law.

### 2.3 First-order Domain-Benchmark Synergy

We now modify the data term so that *different training domains can reduce loss at different rates* on each benchmark, as illustrated in Figure 2. Recall that $u_{i,k} \in [0, 1]$ is the fraction of tokens from training domain $D_k$ that model $M_i$ is pretrained on. It is straightforward to show that:

$$\log d_i = \sum_k u_{i,k} \log(u_{i,k} d_i) + H(u_i), \tag{3}$$

where $H(u_i) = -\sum_k u_{i,k} \log u_{i,k}$. We introduce domain-specific data scaling exponent parameters $\{\gamma_{j,k}\}$, which modify the data scaling exponents per-task additively: $\beta_j + \gamma_{j,k}$. A positive $\gamma_{j,k}$ means that tokens from domain $k$ reduce loss on benchmark $j$ faster than the baseline rate $\beta_j$ predicts, which we refer to as a *synergistic* effect. Conversely, a negative $\gamma_{j,k}$ implies *interference*: tokens from domain $k$ reduce loss more slowly than expected. Substituting (3) into the third argument of the LSE in (2) and introducing these modifiers we have:

$$\mathbb{E}[\, l_{i,j}\,] = \mathrm{LSE}\Big(e_j, a_j - \alpha_j s_i, b_j - \big[\beta_j \mathbf{1} + \gamma_{j,\cdot}\big]^\top z_i - \beta_j H(u_i)\Big), \tag{4}$$

with $z_{i,k} = u_{i,k} \log(u_{i,k} d_i)$ and synergy coefficients $\gamma_{j,k} \in \mathbb{R}$. Equivalently, in the original (non-log) parameterization, this is a multiplier to the data term in (1),

$$L_j(N_i, d_i, u_i) = L_{\infty,j} + A_j N_i^{-\alpha_j} + B_j\, d_i^{-\beta_j} \prod_k (u_{i,k} d_i)^{-\gamma_{j,k} u_{i,k}}, \tag{5}$$

so each domain $k$ rescales the effective data term by a factor $(u_{i,k} d_i)^{-\gamma_{j,k} u_{i,k}}$ that depends on both its token share $u_{i,k}$ and its absolute token count $u_{i,k} d_i$.

### 2.4 Second-order Pretraining Data Synergy

We hypothesize that certain training domains interact synergistically, such that their joint presence yields learning benefits independent of the specific downstream benchmark. These synergies reflect fundamental complementarities between domains, though their effect size is still mediated by the data scaling coefficient of each benchmark. The gain from such co-occurrence is bottlenecked by the scarcer domain and can be understood as producing "additional bonus tokens" only when both domains are present. To capture this, we model synergy with a pairwise term that vanishes if either domain is absent and scales with the smaller per-domain log-token budget.

Define

$$z_{i,k} = u_{i,k} \log(u_{i,k} d_i), \quad \bar{z}_{i,k} = \log\big(1 + u_{i,k} d_i\big), \quad \mathrm{softmin}_\tau(a, b) := -\tau \log\big(e^{-a/\tau} + e^{-b/\tau}\big).$$

Start from the first-order data term split into baseline and per-domain parts, $-\beta_j \sum_k z_{i,k} - \sum_k \gamma_{j,k}\, z_{i,k} - \beta_j H(u_i)$, and modify only the first part with a co-occurrence term

$$-\beta_j \sum_k \Big[ z_{i,k} + \sum_{k' \neq k} \gamma_{j,k} \sigma_{kk'}\ \mathrm{softmin}_\tau\big(\bar{z}_{i,k}, \bar{z}_{i,k'}\big)\Big] \;-\; \sum_k \gamma_{j,k} z_{i,k} \;-\; \beta_j H(u_i),$$

which is a sum over pairwise synergies $\sigma_{kk'}\ \mathrm{softmin}_\tau\big(\bar{z}_{i,k}, \bar{z}_{i,k'}\big)$ modulated by the domain-benchmark synergy strength.

Symmetrizing (using $\sigma_{kk'} = \sigma_{k'k}$) gives

$$-\sum_k (\beta_j + \gamma_{j,k})\, z_{i,k} \;-\; \beta_j \sum_{k<k'} (\gamma_{j,k} + \gamma_{j,k'})\, \sigma_{kk'}\ \mathrm{softmin}_\tau\big(\bar{z}_{i,k}, \bar{z}_{i,k'}\big) \;-\; \beta_j H(u_i).$$

Using $\bar{z}$ (nonnegative and $= 0$ when $u_{i,k} = 0$) ensures that the interaction truly vanishes if either domain is absent and preserves the desired $O(\log d_i)$ scaling. The pairwise term carries the same $\beta_j$ as the baseline, so it moves at the same rate and reads as a change in *effective data count*. Without the $\beta_j$ factor, $\sigma_{kk'}$ would be harder to interpret across benchmarks with different $\beta_j$.

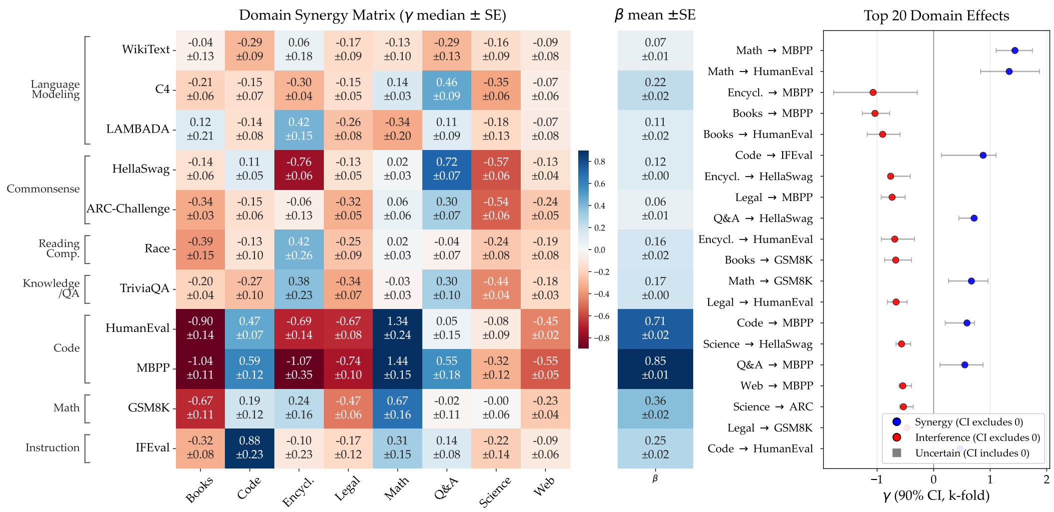


Figure 3: **Overview of estimated domain→benchmark synergy.** We fit domain-aware scaling laws across open-weight LLMs and extract per-domain synergy coefficients $\gamma_{j,k}$ that quantify how each pretraining domain contributes to benchmark performance beyond aggregate token count. ***Left:*** Heatmap of $\gamma_{j,k}$ (median $\pm$ SE) between pretraining domains (columns) and benchmarks (rows), where positive values show synergy and negative values visualize interference. ***Center:*** Benchmark-specific data exponent $\beta_j$. ***Right:*** Top 20 first-order synergies ranked by magnitude, with 90% bootstrap confidence intervals.

**Formulation.** Starting from the LSE formulation of the baseline scaling law, plugging in the new data term we have

$$\Phi_{i,j} = b_j - \left[\beta_j \mathbf{1}_K + \gamma_{j,\cdot}\right]^\top z_i - \beta_j\, H(u_i) - \beta_j \sum_{k<k'} (\gamma_{j,k} + \gamma_{j,k'})\, \sigma_{kk'}\ \operatorname{softmin}_\tau(\bar{z}_{i,k}, \bar{z}_{i,k'}), \quad (6)$$

where $\Sigma = [\sigma_{kk'}]$ is symmetric with $\sigma_{kk} = 0$. The complete log-space law has the form

$$\log L_{i,j} = \operatorname{LSE}\big(e_j,\ a_j - \alpha_j s_i,\ \Phi_{i,j}\big).$$

Intuitively, the pairwise term can be understood as producing benchmark-specific *effective tokens*. Define $\log D^{\text{eff}}_{i,j} := \log d_i + \sum_{k<k'} (\gamma_{j,k} + \gamma_{j,k'})\, \sigma_{kk'}\, s_{i,kk'}$, where $s_{i,kk'} = \operatorname{softmin}_\tau(\bar{z}_{i,k}, \bar{z}_{i,k'})$. Reordering the terms to achieve a view similar to Chinchilla, the data term scales with $(D^{\text{eff}}_{i,j})^{-\beta_j}$. This shows that the co-occurrence of synergistic domains contributes "bonus tokens" that increase the effective dataset size for a given benchmark, while interfering pairs reduce it. A complete derivation is provided in Appendix A.

## 3 Estimation from Observational Data

A distinguishing feature of our approach is that all scaling law parameters, including domain synergies, are estimated from *observational* data, a collection of publicly available pretrained language models with documented pretraining mixtures, rather than models trained in controlled experiments. Because the data is observational rather than interventional, the synergy coefficients we estimate should be understood as associations, i.e., they capture how data composition varies with benchmark performance across the observed model set, but do not by themselves establish causal domain interactions. We take a step toward causal validation in Section 4.3. This broad observational coverage comes with several estimation challenges, which we address through normalization, regularization, cross-validation, and uncertainty quantification. We summarize the data, evaluation choices, and fitting methodology below, and details are deferred to Appendices B, C, and D.

**Data.** Our model set spans six open-weight model groups (`GPT-Neo/J/NeoX` (Black et al., 2021; Wang & Komatsuzaki, 2021; Black et al., 2022), `Pythia` (Biderman et al., 2023), `DataDecide` (Magnusson et al., 2025), `OLMo` (Groeneveld et al., 2024), `OpenLLaMA` (Geng &

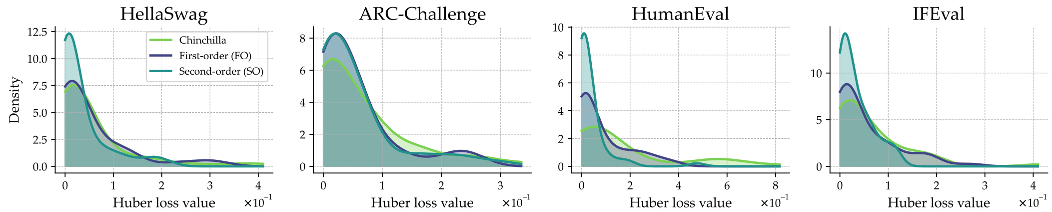


Figure 4: **Distribution of held-out prediction loss for three fitted scaling models**: We evaluate the Huber loss between each fitted scaling law's prediction and the observed value on held-out test examples, for the domain-agnostic Chinchilla baseline and the first-order (FO) and second-order (SO) synergy models, across four benchmarks. Both synergy-aware models concentrate more density near zero, which means that they achieve more accurate predictions on unseen models. The largest improvements appear on `HumanEval` and `IFEval`.

Liu, 2023), `RedPajama-INCITE` (Weber et al., 2024)), 52 models in total from 70M to 20B parameters. We map each model's pretraining sources to eight domains (*books*, *code*, *encyclopedia*, *legal*, *math*, *Q&A*, *science*, *web*). These domains are aligned with the Data Provenance Initiative (Longpre et al., 2023) and Essential-Web v1.0 (Essential AI, 2025) data categories. The 30 `DataDecide` models add controlled pretraining data ablations (*no-code*, *no-math-code*, *no-flan*, *no-reddit*) that are especially informative for estimating synergies. We evaluate on 11 benchmarks: `HumanEval` (Chen et al., 2021), `MBPP` (Austin et al., 2021), `GSM8K` (Cobbe et al., 2021), `ARC-Challenge` (Clark et al., 2018), `HellaSwag` (Zellers et al., 2019), `RACE` (Lai et al., 2017), `TriviaQA` (Joshi et al., 2017), `LAMBADA` (Paperno et al., 2016), `IFEval` (Zhou et al., 2023), `WikiText` (Merity et al., 2016), and `C4` (Raffel et al., 2019). Because many models are small, exact match scores on open-ended tasks are often near zero, so we score those tasks by bits-per-byte (BPB) of the gold response and rank-Gaussian transform (Conover & Iman, 1981) all targets to a common scale before fitting.

**Identifiability.** In observational data, $\log D$ and $\log N$ are correlated, so the Chinchilla exponents $\alpha$ and $\beta$ trade off freely. However, the composition features $z_{i,k} = u_{i,k} \log(u_{i,k} d_i)$ are nearly orthogonal to $\log N_i$, which means the synergy coefficients $\gamma$ are well-identified even when $\alpha$ and $\beta$ individually are not. An important caveat is that the observed mixture vectors are effectively low-dimensional, so we interpret $\gamma_{j,k}$ as composition sensitivities along the observed mixture axes rather than as fully causal synergy estimates (Appendix D).

**Robust fitting.** We fit both the first-order (FO) and second-order (SO) models in two stages. First, we fit a per-task Chinchilla baseline (1), then the full synergy model. We place a prior on each $\beta_j$ centered at its Chinchilla value $\hat{\beta}_j^{\text{chin}}$, which keeps the data scaling exponent positive and lets the synergy terms capture composition, so the fit stays interpretable at no cost in accuracy. Fitting the non-convex LSE objective on a small number of models is prone to overfitting, so we use a Huber loss (Huber, 1992) for robustness and $\ell_1$ and $\ell_2$ regularization to make the $\gamma_{j,k}$ sparse under collinear domain fractions. We minimize the loss using the L-BFGS algorithm (Nocedal, 1980).

**Uncertainty and model selection.** We evaluate the fit by 5-fold cross-validation over the models and quantify uncertainty by refitting on $B{=}50$ bootstrap resamples (80% of models). We consider only synergy entries whose 90% intervals exclude zero as meaningful. The FO model is fit per task, while the SO model is fit jointly since $\Sigma$ is shared across tasks. We select fits by *parameter stability*, the consistency of $\gamma$ across folds, alongside held-out $R^2$ (Appendix B).

## 4 Results

We present our findings in three parts. First, we estimate the first-order domain→benchmark synergies (Section 4.1). Second, we estimate the second-order pairwise interactions between pretraining domains (Section 4.2). Finally, we validate these results through controlled mixture-optimization experiments (Section 4.3).

### 4.1 First-order Domain→Benchmark Synergy

We first estimate the first-order synergy of each pretraining domain on each benchmark, which captures how each domain contributes to benchmark performance beyond aggregate token count. Our first-order (FO) domain-aware model adds a domain coefficient $\gamma_{j,k}$ to each benchmark's data scaling exponent, $\beta_j + \gamma_{j,k}$ (4): a positive $\gamma_{j,k}$ means tokens from domain $k$ increase the scaling rate on benchmark $j$ above its baseline $\beta_j$. We fit this model to the observational data in Section 3.

**Strongest synergies.** Figure 3 shows the estimated matrix $\Gamma = \{\gamma_{j,k}\}$, the data scaling exponents $\beta_j$, and the top 20 synergies ranked by magnitude, each with bootstrap standard errors and 90% confidence intervals. The strongest synergies appear for code and math benchmarks. For example, `HumanEval` shows $\gamma_{\text{code}} = +0.47$ and $\gamma_{\text{math}} = +1.34$, and `MBPP` shows $\gamma_{\text{code}} = +0.59$ and $\gamma_{\text{math}} = +1.44$. These values are consistent with prior findings that math and code data reinforce each other on programming benchmarks (Azerbayev et al., 2023; Lu et al., 2024). The synergies are robust across cross-validation folds and sign-consistent for most benchmark-domain pairs (Appendix B), and the same fit also identifies reliable interference, most clearly from books and encyclopedia on code benchmarks. Composition affects not only the benchmark performance but the rate at which it scales with data from each domain. Figure 2 shows the data exponent $\beta$ on `HumanEval` falls from 0.45 under the full `dolma1.7` mixture to 0.18 once code is removed (Appendix E.2).

**First-order fit.** The FO model obtains a median cross-validated $R^2$ of 0.906 across benchmarks. Its gain over the domain-agnostic Chinchilla baseline is largest on the code benchmarks, where the total token count alone explains little of the variance: the baseline reaches only $R^2 = 0.41$ on `HumanEval` and 0.20 on `MBPP`, while the FO model reaches 0.92 and 0.88 (Appendix Figure 8).

### 4.2 Second-order Pretraining Data Synergy

Beyond first-order effects, we estimate *second-order* data synergy between pretraining domains, which captures gains that materialize only when two domains co-occur. The SO model introduces shared pairwise domain interaction parameters $\sigma_{kk'}$, estimated jointly across all benchmarks. We fit the symmetric synergy matrix $\Sigma = \{\sigma_{kk'}\}$ along with other parameters as in (6) and evaluate uncertainty by bootstrap resampling.

**Strongest pairwise synergies.** Figure 5 summarizes the shared pairwise synergy matrix $\Sigma = \{\sigma_{kk'}\}$ across domains, along with the confidence intervals. The largest pairwise interactions are $\sigma_{\text{code,science}} = +2.55$ and $\sigma_{\text{code,math}} = +2.40$. When code co-occurs with math or scientific data in the pretraining mixture, the pair contributes more than the sum of the two domains' first-order effects. The clearest negative interaction is *Math*×*Books*: their co-presence in the pretraining corpus may lower average performance on the affected benchmarks. This suggests that the two domains share little structure, so tokens spent on both dilute each other rather than provide the mutual reinforcement seen for code and math. A negative $\sigma$ is therefore actionable, when a target benchmark favors one such domain, a mixture should not also spend tokens on its interfering domains.

**Second-order fit.** The SO model obtains a median cross-validated $R^2$ of 0.912 across benchmarks and improves on FO most where FO already showed strong domain effects: `GSM8K` (+0.013), `IFEval` (+0.020), and `TriviaQA` (+0.019). Its per-benchmark $\gamma$ coefficients shrink relative to FO (Appendix Figure 12), because the shared $\sigma$ absorbs part of the domain→benchmark synergy. On held-out prediction loss (Figure 4), both the FO and SO models place more of the loss distribution near zero than the domain-agnostic baseline, so the fitted $\gamma_{j,k}$ and $\sigma_{kk'}$ generalize rather than overfit, most clearly on benchmarks unlike typical pretraining text such as `HumanEval` (Appendix E.1 reports all fits). Additionally, our first- and second-order laws also achieve the best held-out fit among prior composition-aware scaling laws (Appendix F).

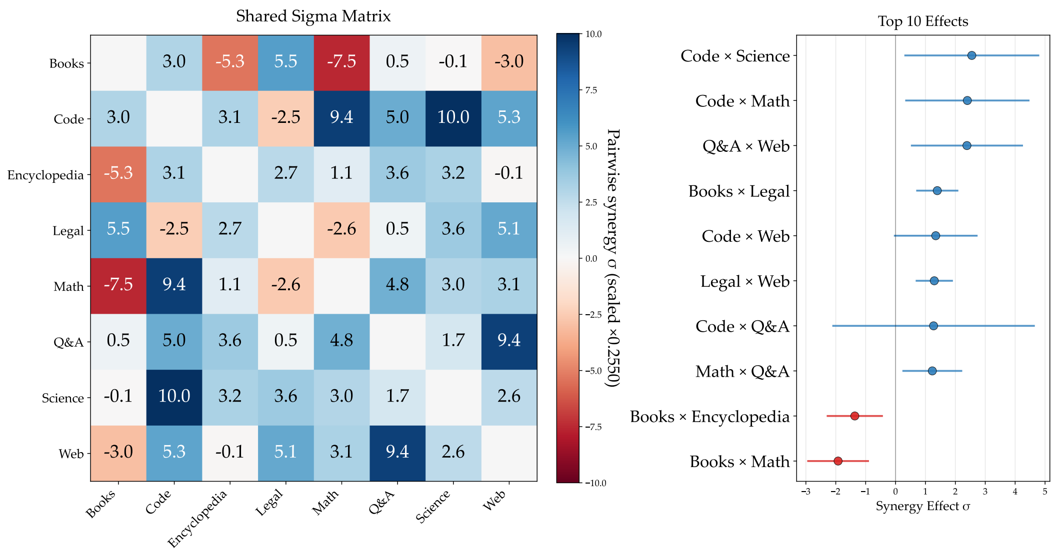


Figure 5: **Second-order domain-domain synergy.** ***Left:*** The shared pairwise interaction matrix $\Sigma = \{\sigma_{kk'}\}$, where positive entries show synergy between two domains and negative entries show interference. ***Right:*** The largest second-order synergies with 90% bootstrap confidence intervals. Code shows the strongest positive synergy with math and science data, which appears only when the domains appear together in pretraining data.

### 4.3 Validation via Mixture Optimization

The synergy estimates above are observational. To test whether they are *actionable*, we use the domain-aware scaling laws along with estimated first-order synergies to predict benchmark performance, and leverage those predictions for pretraining mixture optimization. We train models at two scales, 30M and 150M parameters, on mixtures predicted to be optimal or anti-optimal for three target tasks: `HumanEval`, `GSM8K`, and `IFEval`. Appendix G.1 shows that the predicted-optimal mixture puts the most weight on the domains with the largest positive $\gamma_{j,k}$ (e.g., *math* and *code* for `HumanEval`) and the least on the rest, and the anti-optimal mixture does the reverse (e.g., *books* and *encyclopedia*). We optimize over the domains from `DataDecide`, and the balanced baseline gives equal weight to each domain. Full details of this experiment are in Appendix G.2.

Table 1: Bits-per-byte (BPB, ↓) on target tasks at two model scales. The scaling law correctly predicts the ranking between optimal and anti-optimal for all tasks at both scales. Percentages are relative to the balanced baseline.

| Metric (BPB) | Balanced | Optimal | Anti-optimal | Gap |
|---|---|---|---|---|
| *30M parameters, 3B tokens* | | | | |
| `HumanEval` | 1.268 | **1.110** (−12.4%) | 1.503 (+18.6%) | 0.393 |
| `GSM8K` | 1.818 | **1.779** (−2.1%) | 1.935 (+6.5%) | 0.156 |
| `IFEval` | **1.451** | 1.513 (+4.3%) | 1.551 (+6.9%) | 0.038 |
| *150M parameters, 5B tokens* | | | | |
| `HumanEval` | 0.868 | **0.721** (−16.9%) | 1.058 (+21.9%) | 0.337 |
| `GSM8K` | 1.643 | **1.402** (−14.7%) | 1.591 (−3.2%) | 0.189 |
| `IFEval` | 1.366 | **1.324** (−3.0%) | 1.365 (−0.0%) | 0.041 |

Table 1 reports bits-per-byte on each task's held-out set. At 30M, the predicted-optimal mixture outperforms the anti-optimal on all three tasks, which confirms the scaling law's directional prediction. The gap is largest on `HumanEval`, where the optimal mixture obtains a 31% BPB improvement (relative to the balanced baseline) over the anti-optimal.

At 150M, the predictions still hold for all three tasks, and the relative gaps widen: `HumanEval` keeps a large gap (0.337 BPB), `GSM8K` strengthens from a 2.1% to a 14.7% reduction over balanced, and `IFEval` stays small but consistent. The synergy coefficients $\gamma_{j,k}$, estimated purely from observational data, thus carry actionable insight for mixture curation that persists as model scale increases.

## 5 Related Work

**Scaling laws.** LLM pretraining validation loss follows power laws as model capacity and data grow. Early results characterize scaling with parameters, data, and compute (Kaplan et al., 2020), while later work refines the compute-data trade-off and recommends that token counts scale roughly with parameters to remain compute-optimal (Hoffmann et al., 2022). Recent work fits such laws *observationally*, from existing public models: a low-dimensional capability space predicts downstream performance (Ruan et al., 2024), and fitted parameters transfer across models that differ in architecture and data (Choshen et al., 2024). We treat these domain-agnostic laws as our baseline and add terms that depend on data composition to scaling laws.

**Domain-aware scaling and mixture optimization.** Beyond total token count, several works show *which* tokens matter. Data pruning can reduce error to exponential decay in dataset size given a good pruning metric (Sorscher et al., 2022). Mixture optimization methods select the domain mixture that minimizes loss through regression (Liu et al., 2024) or parametric mixing laws that capture nonlinear returns (Ye et al., 2024), and online methods further adapt the mixture during training through unified reweighting (Chen et al., 2024a) or gradient-based balancing (Ge et al., 2025). Closest to our work, a few methods fit variations of scaling laws to select the mixture (Kang et al., 2024; Shukor et al., 2025; Longpre et al., 2025), but they keep the data scaling exponent global or use additive per-domain power laws instead of modifying the data term. Our first-order model instead embeds composition into the scaling exponent itself ($\beta_j + \gamma_{j,k}$, which varies by both domain and benchmark), so that mixture changes modify the *rate* at which additional data reduces the loss. Our second-order extension further introduces explicit pairwise domain interactions ($\sigma_{kk'}$) that none of these methods capture.

**Observational inference of skills and performance.** A complementary literature utilizes observational variation across many (model, task) points: perplexity correlations guide data selection without training (Thrush et al., 2024), latent variable models recover shared capability factors (Jin et al., 2025), and data influence methods show that group interactions can cancel or amplify individual effects (Yu et al., 2025). We share this spirit but predict performance under counterfactual mixtures from domain-level composition. Other related work predicts downstream performance from pretraining loss or compute (Chen et al., 2024b; Schaeffer et al., 2025), including on generative benchmarks.

**Evidence for data synergy.** Multiple empirical studies report positive transfer between code and mathematical reasoning. Continued pretraining that combines math and code improves math benchmarks (Azerbayev et al., 2023; Lu et al., 2024; Wang et al., 2026), code during pretraining rather than only at the SFT stage improves broader reasoning with little negative transfer (Ma et al., 2023), and code pretraining improves entity tracking (Kim et al., 2024). These findings motivate sparse, domain-specific parameters that capture synergy between conceptually related domains.

## 6 Conclusion & Discussion

We formalize and quantify data synergy in LLM pretraining and show that explicitly modeling domain-benchmark and second-order synergy between training domains recovers interpretable, sparse synergy estimates and improves predictive fit across multiple benchmarks. We also note that one could discover synergies by changing the definition of domains, e.g., different modalities, different languages, or finer splits of code and science. We estimate synergy in pretraining, but whether the same domain pairs stay synergistic under supervised fine-tuning or reinforcement learning remains open.

Our approach is limited by its observational design, as estimates rely on potentially noisy or incomplete mixture metadata, and the effective mixture space is low-dimensional. The validation experiments remain at small scale, so verification at larger model sizes is an important direction for future work. Beyond methodology, our synergy estimates support practical applications in data curation, mixture design, and data acquisition, and offer a principled tool for targeting specific capabilities in future training runs.